

Source Scaling in B2-Eirene – Basics

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The problem arises from a several orders of magnitude difference between the recycling neutral sources in the divertor plasma and the neutral particle throughput in the pumping and fuelling system. If the walls are saturated and no cryogenic pump is present directly in the divertor chamber, then the particle throughput becomes one of very few means for controlling the plasma density in the discharge. Calculations for ITER [1] yield the throughput at the level of 10^{23} particles per second, whereas the total strength of ionisation sources in the plasma due to neutral recycling on the material surfaces and volumetric recombination of the plasma exceed 10^{25} particles per second. The latter sources determine the scale of the source terms in the plasma transport equations, hence if a resolution of the order of one per cent in the throughput is required (and this is a usual requirement in the density scans for ITER), then the relative accuracy of the particle balance in the coupled B2-Eirene run must be better than 10^{-4} , whereas a direct application of the B2-Eirene code results in the accuracy of 10^{-2} at best. Therefore, a special effort to reduce the error in the particle balance in the coupled fluid-Monte-Carlo code, such as the B2-Eirene package, is required.

The sources of the errors in the particle balance attributed to the code coupling procedure can be separated in two groups. First, the Eirene (at least, the “old” one) preserves the accurate particle balance in each stratum only for the totals of all the particle species in this stratum, e. g., compensating for under-estimated D source by over-estimating the He source (one goose plus one horse). Second, the coupling is done in a combined, explicit-implicit way, where the ion sources due to the neutral ionisation generally have both explicit and implicit components, and the plasma fluxes to the walls are implicit. This leads to a violation of global particle balance during each time step for the fluid equations. Consider, for example, the particle balance during recycling on a single target. Let Γ_{div} be the ion flux to the target, Γ_{pmp} the neutral flux to the pump, and S the total ionisation source in the plasma. Then the neutral recycling flux $I = \Gamma_{div} - \Gamma_{pmp}$ and the ion recycling flux S are equal after each call to Eirene, that is, before a B2 step, and they evolve differently during the B2 iterations. Indeed, Γ_{div} is the integral of the ion flow over the target, which depends on the local ion density, and S is an integral over the plasma volume of a source which has an implicit part (i.e. proportional to the ion density). The resulting discrepancy is normally within a few percent of the source, but this is still bigger than the pumped flux.

Following this classification, the following procedure is proposed (and realised in the B2-Eirene/solps4.0). First, the ionisation sources and the neutral flows to the surfaces are modified after every call to the Eirene in order to ensure an accurate particle balance in each stratum for each plasma species separately. Second, scaling factors ~ 1 are introduced for each neutral source and are adjusted in course of the B2 iterations so that the neutral and ion recycling fluxes remain in balance.

1. Initial source scaling.

Consider the particle balance in a single stratum for a single species. Let Γ_{pi} be the ion flux to the recycling surfaces, Γ_{po} the flux of resulting neutrals from the surfaces (we count indeed the nuclei, so that one D_2 molecules counts twice), Γ_{ni} the neutral flux onto the surfaces, Γ_{no} the resulting neutral flux back from the surfaces, Γ_g the flux of neutrals introduced with gas puffing, Γ_s the equivalent flux of neutrals lost in the neutral-plasma interactions (negative for recombination), and Γ_t the equivalent neutral flux lost in the “time layer”. All these quantities are calculated in Eirene, and the ion flux on the recycling surfaces, as calculated in B2, is Γ_{B2} . Note that Γ_s is the contribution of this stratum to the ion sources in Eirene, and the difference $(\Gamma_{ni} - \Gamma_{no})$ normally represents the flux to the pump and to the plasma core.

(a) Recycling stratum.

First, make sure that the ion flux to the surfaces as seen from the B2 and Eirene sides is the same. For this, multiply all the fluxes (Γ_{po} , Γ_{ni} , Γ_{no} , Γ_s , Γ_t) by the ratio of Γ_{B2}/Γ_{pi} . Then find a factor ξ from the particle balance equation

$$(\Gamma_s + \Gamma_t + (\Gamma_{ni} - \Gamma_{no})) \xi = \Gamma_{po} \quad (1)$$

and multiply Γ_s , Γ_t , Γ_{ni} , and Γ_{no} by ξ .

(b) Non-recycling stratum (recombination or gas puff)

The equation for the correction factor ξ is here

$$(\Gamma_s + \Gamma_t) \xi + (\Gamma_{ni} - \Gamma_{no})/\xi = \Gamma_g \quad (2)$$

and, correspondingly, Γ_s and Γ_t are multiplied, and Γ_{ni} and Γ_{no} are divided, by ξ . From the two solutions of Eq. (2), the positive one closest to 1 should be chosen.

The intuitive rationale for different form of the equations for recycling and non-recycling strata is that in case of recombination, Γ_s is negative and the correction factor is closer to 1 if it is applied to the positive and negative terms in different directions. Probably, one should solve both the linear and

quadratic equations for non-recombining strata, selecting afterwards the closest to 1 positive solution. I_t is typically small and should not play a significant role.

After this first phase of correction, the integral particle balance in each stratum for each species before the B2 time step should be accurate to the machine precision ($\sim 10^{-15}$).

2. Source scaling during the B2 iterations.

Again, consider the particle balance in a single stratum for a single species. Let I^* be the neutral source, and S^* be the ion source after the first correction, where they are in balance. The I^* is I_{po} for a recycling stratum, I_g for a gas-puffing stratum, or a volume integral of negative sources contributing to I_s for a recombination stratum. The S^* is a volume integral of positive sources contributing to I_s for a recombination stratum or I_s for the rest. Such a split in recombination stratum is motivated by a sharp temperature dependence of the recombination and ionisation rates in the transition from recombination-dominated to ionisation-dominated regions, so that equally strong recombination and ionisation in one cell are rather unlikely.

We introduce two scaling factors, α and λ , and scale the B2 sources and neutral fluxes for diagnostics in the following way

$$S(x,y) = \alpha S^*(x,y) \quad (3)$$

$$S_m(x,y) = \lambda S_m^*(x,y), S_E(x,y) = \lambda S_E^*(x,y), j_w = \lambda j_w^* \quad (4)$$

Here S_m stands for momentum sources, S_E for energy sources, and j_w for the neutral fluxes on the surfaces, which are needed, first of all, to calculate the pumping rate. Now the neutral and plasma sources after the i -th sub-iteration in B2 are $I(\alpha,i)$ and $S(\alpha,i)$, and the equation for α is

$$\frac{S^*}{I^*} \frac{I(\alpha,i)}{S(\alpha,i)} = 1 \quad (5)$$

– that is, the fraction of neutrals ionized in the plasma in course of the sub-iterations does not change. This equation is solved during the same sub-iterations, with a simple iteration scheme

$$\alpha_{i+1} = \frac{S^*}{I^*} \frac{I(\alpha,i)}{S(\alpha,i)} \alpha_i \quad (6)$$

The factor λ can be defined as

$$\lambda_i = I(\alpha,i)/I^*, \quad (7)$$

that is, the momentum and energy sources, together with neutral fluxes, are proportional to the neutral source.

- [1] A. S. Kukushkin, H. D. Pacher, Plasma Phys. Control. Fusion **44** (2002) 931